



Between Whispers and Screams

Loudness Standard Deviation as a Proxy for
Explicit Content Detection in Romance Fiction

Svenja Guhr



Sound in Fiction

Soundscape
=
sound + landscape



pixabay

(see Schafer 1994)

Sound in Fiction

“ Rachel was lost, floating, adrift with the universe, tied forever with the ocean and the land and the man in her arms, who was bringing her back to the edge of fulfillment and taking her beyond time [...].

Until she was weeping against his sweat-slick shoulder, clinging to him helplessly. "Don't leave me," she whispered brokenly against the damp skin. "Come with me."

And when the next wave hit her, she felt his body tense in her arms, heard his muffled cry, and they were there together, lost and found, flung out into the gently rocking ocean of eternity.

(Anne Stuart 1984, *Tangled Lies*)



Sound in Fiction

"Don't leave me," `<character_sound loudness="2.0">`she whispered brokenly against the damp skin`</character_sound>`.
"Come with me."

The wind around the cottage was picking up,
`<ambient_sound loudness="3.5">`the waves had crescendoed to a muffled roar`</ambient_sound>`

(Anne Stuart 1984, *Tangled Lies*)

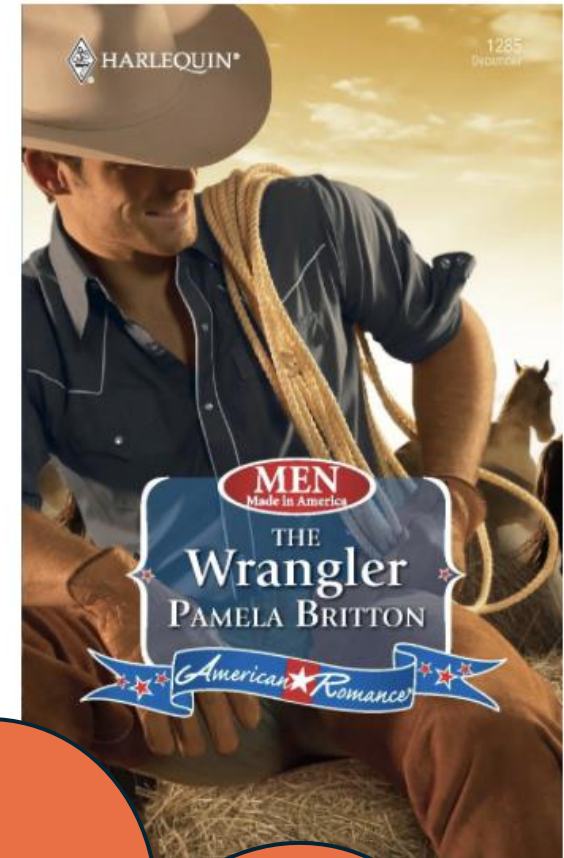
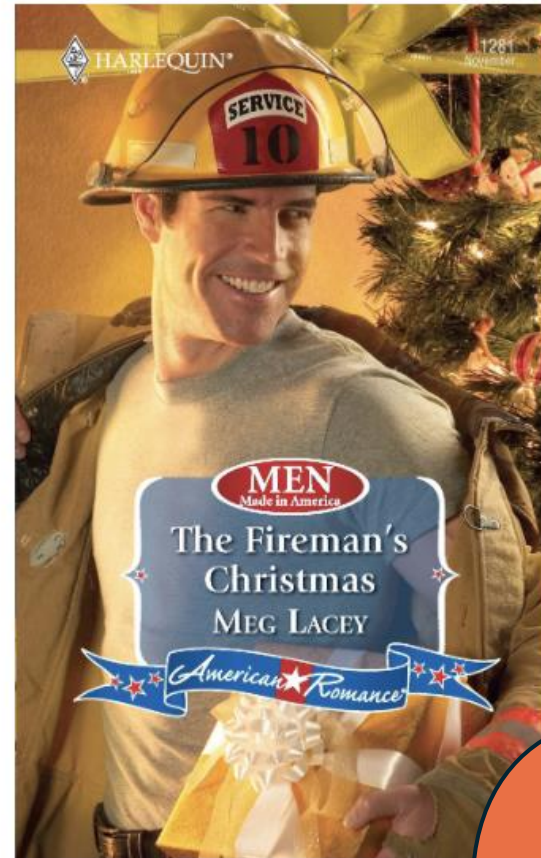
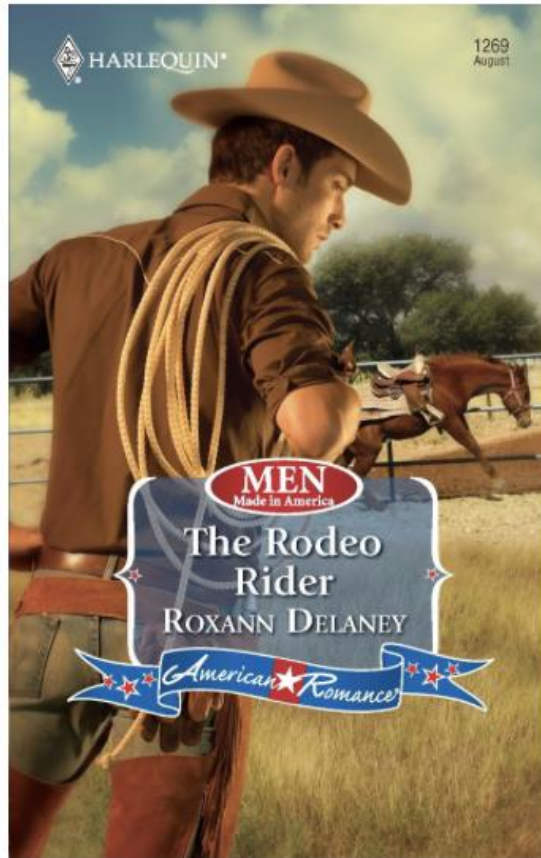
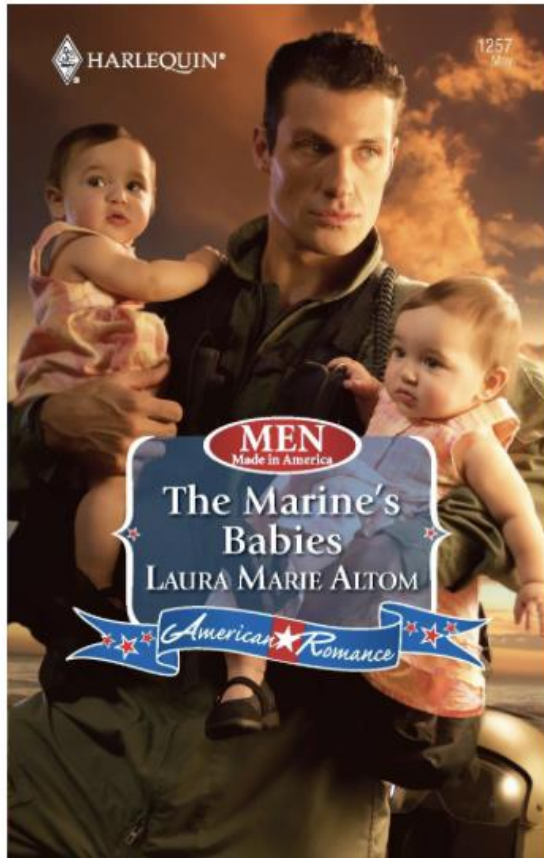


Research Question and Hypotheses

- **Q:** To what extent does loudness SD serve as a proxy for identifying *violent/scary or erotic* scenes?

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- **Q:** To what extent does loudness SD serve as a proxy for identifying *violent/scary or erotic* scenes?
- **H1:** Scenes annotated as *violent/scary* will show significantly higher *ambient sound* loudness SD than *other* scenes, reflecting the dynamic contrast between loud events and silence characteristic of threat.
- **H2:** Scenes annotated as *erotic* will show significantly higher *character sound* loudness SD, reflecting the dynamic range from whispered dialogue to vocalized arousal.



Corpus: Harlequin's Men Made in America

Thanks to
Hannah Walser
who purchased
this corpus in
2015!

Fictional Sound Events

fictional sound = sound event with asserted *realis* in the fiction

- explicitly represented by sound words in the narrative
- classified into *ambient sounds* and *character sounds*
- assigned loudness levels on a 0; 1–5 scale via dictionary matching

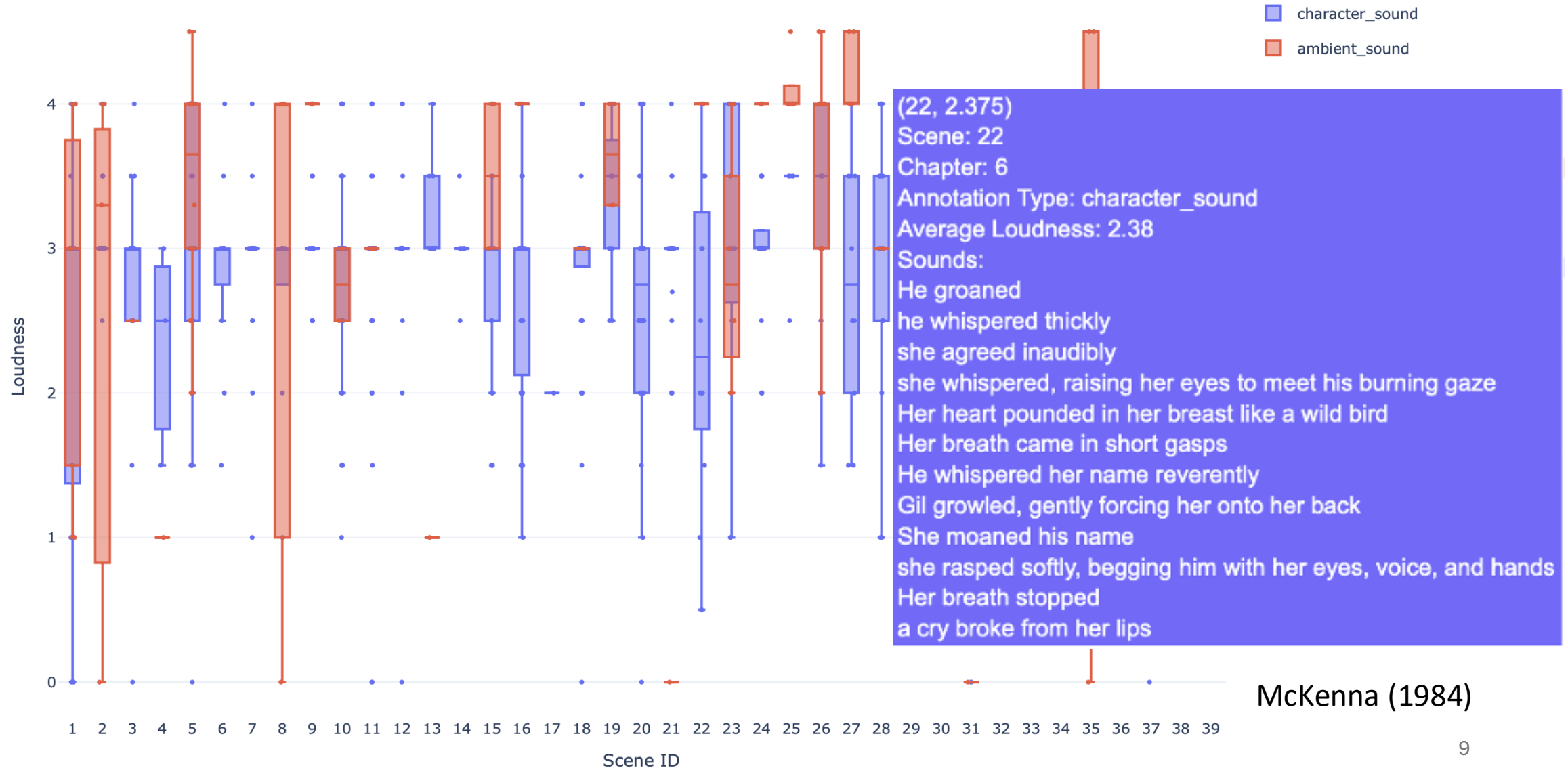
S	explicitly no sound	silence (<i>Schweigen</i> and <i>Stille</i>)
1	very low sound	falling leaves – breathing
2	low sound	ticking of a clock – whispering
3	medium loud sound	singing of a bird – room level conversations
4	loud sound	loudest human-made sound – barking dog
5	very loud sound	car horn – rocket launch

Guhr (2024, 2026)

Loudness Standard Deviation (SD)

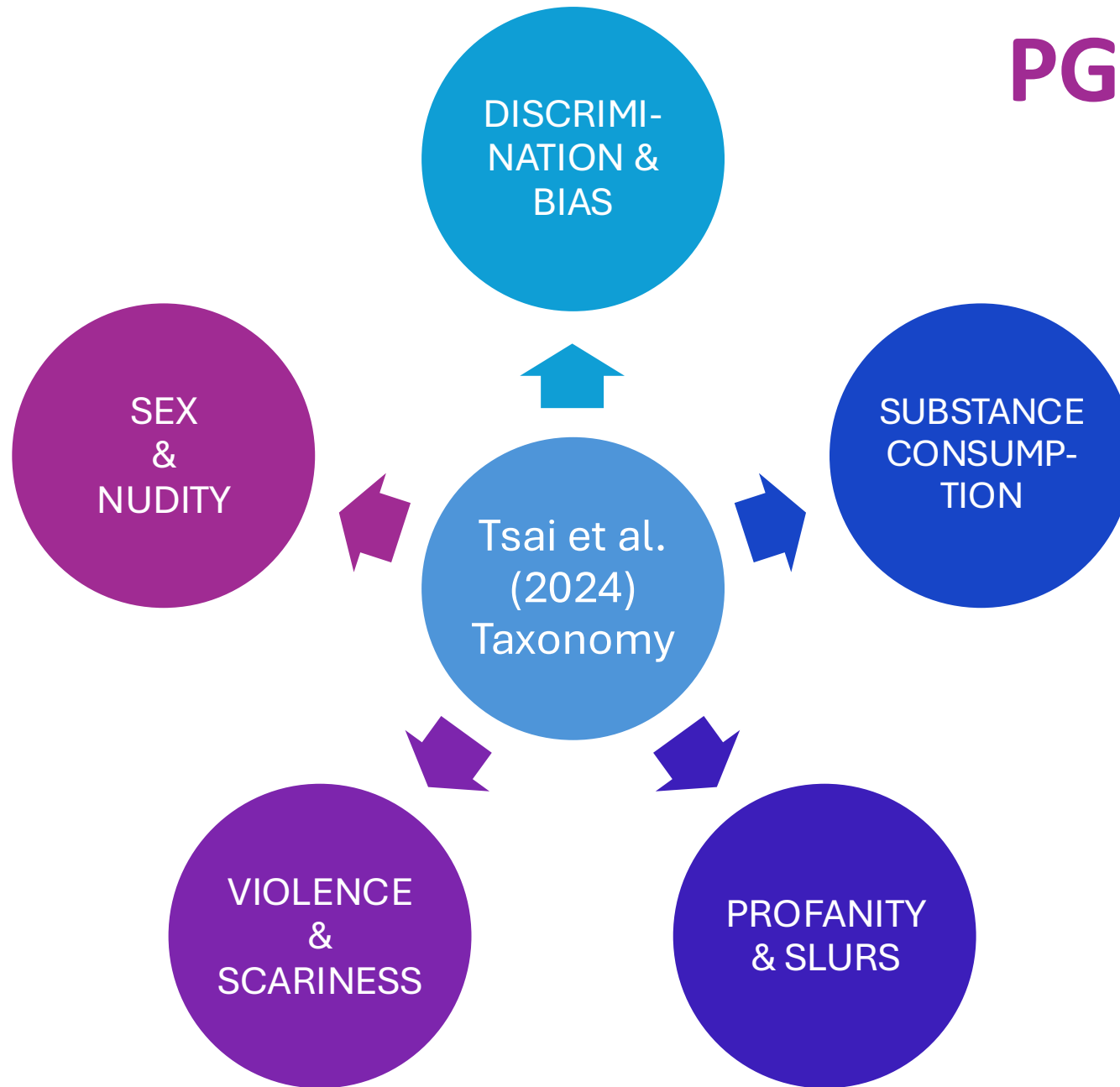
T	text
w	number of words
sw	sound word
se	sound-event span
SW_D	sound-word density
SE_D	sound-event density
T_{se}	number of sound events in a text
T_{pe}	number of possible events in a text
l	loudness-level value
l_{sw}	loudness-level value of a sound word
l_{se}	loudness-level value of a sound event
SW_l	number of loudness level-labeled sound words
SE_l	number of loudness level-labeled sound events
T_{SE_l}	number of loudness level-labeled sound-event spans in a text
$\overline{T_l}$	average loudness level of a text
$\overline{T_{l_A}}$	average loudness level of ambient sounds
$\overline{T_{l_C}}$	average loudness level of character sounds
T_l^{SD}	SD of loudness levels in a text
$T_{l_A}^{SD}$	SD of ambient-sound loudness levels
$T_{l_C}^{SD}$	SD of character-sound loudness levels

Loudness SD in *Too Near the Fire*



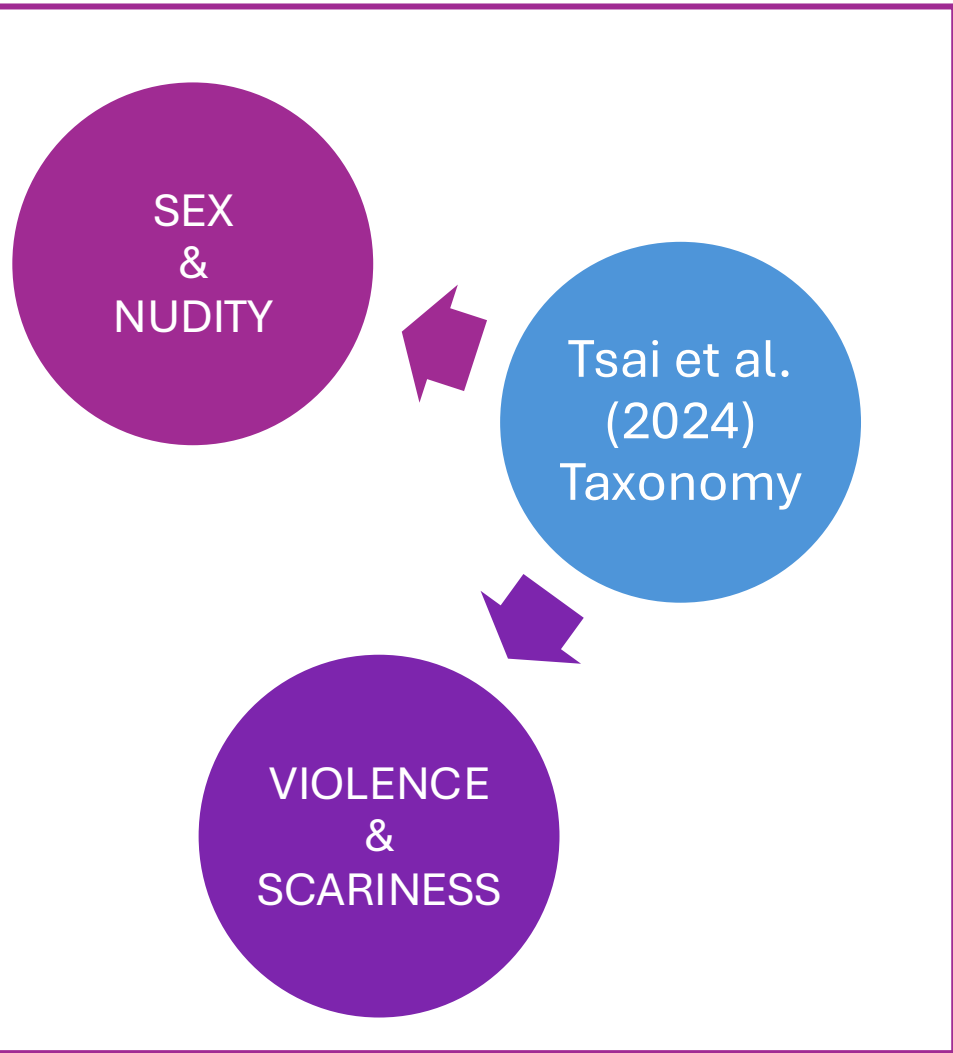
PG-Story Content Safety Taxonomy

absent vs. present
(text classification)



Tsai et al. (2024: 79)

PG-Story Content Safety Taxonomy



absent vs. present
(text classification)



absent / weak / strong
(scene classification)



Tsai et al. (2024: 79)

XLM RoBERTa

- IAA (Cohen's kappa)
 - $\kappa = 0.82$ SEX & NUDITY class
 - $\kappa = 0.79$ VIOLENCE & SCARINESS class

task	class	train	val	total (%)	weight
VIOLENCE/SCARINESS	<i>absent</i>	68	17	85 (34.0%)	0.980
	<i>weak</i>	50	13	63 (25.2%)	1.323
	<i>strong</i>	82	20	102 (40.8%)	0.817
SEX/NUDITY	<i>absent</i>	66	16	82 (32.8%)	1.016
	<i>weak</i>	66	17	83 (33.2%)	1.004
	<i>strong</i>	68	17	85 (34.0%)	0.980
total scenes		200	50	250	

Table 2: Training and validation set statistics for the two classification tasks. Class weights are computed inversely proportional to class frequency to address label imbalance.

External test set: 30 doubly-annotated scenes by 2 independent human annotators:

- The SEX & NUDITY classifier achieved macro F1 = 0.63 on the external test set
- The VIOLENCE & SCARINESS classifier achieved macro F1 = 0.30 on the external test set

LLM-generated synthetic data

- Used prompt: “Please generate [n] passages of max. 350 words each for the class [level] [category] following the operationalization definitions [xy]. Output a csv.”

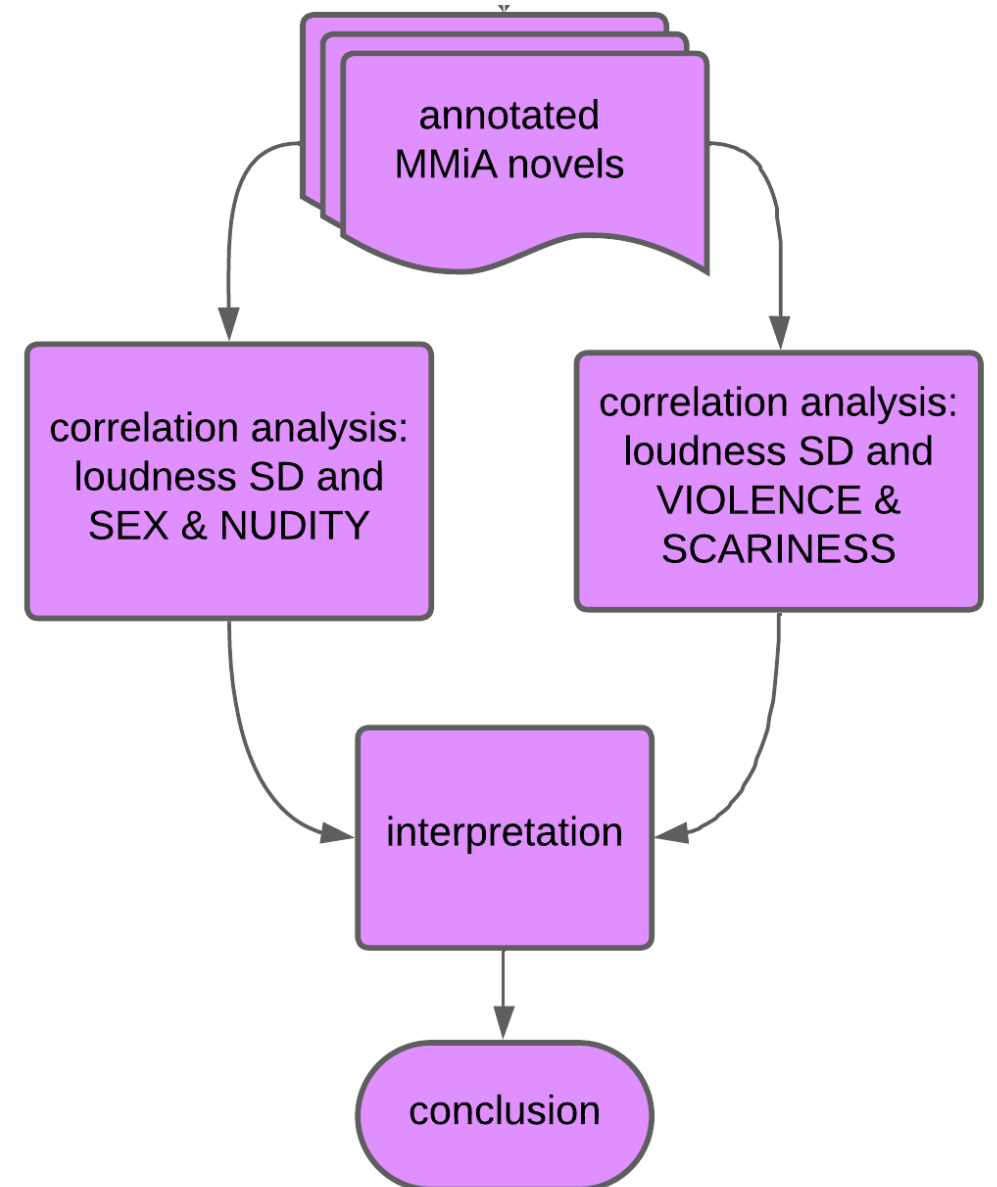
Sample passage generated with Anthropic *claude-sonnet-4-20250514* :

“She arched under his hands and **told him** what she liked and he gave her what she liked with a generosity that made her think she had been very right about him. She came under his hands **loudly** and without apology and he held her through it and entered her in the same motion and she felt the deep pleasure of it.”

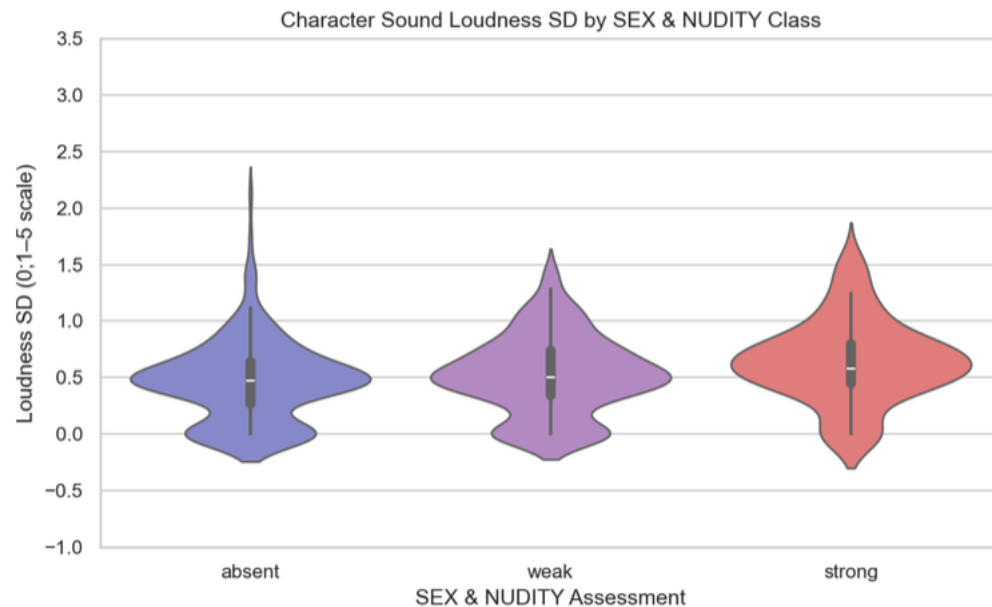
Correlation Analysis

2 Analyses:

- 1) Are *character sound* and *ambient sound* loudness SD associated with *violent/scary* scenes?
- 2) Are *character sound* and *ambient sound* loudness SD associated with *erotic* scenes?



Character Sound Loudness SD



(b) Distribution of *character sound* loudness SD per scene across SEX & NUDITY assessment classes (*absent*, *weak*, *strong*).



(d) Distribution of *character sound* loudness SD per scene across VIOLENCE & SCARINESS assessment classes (*absent*, *weak*, *strong*).

Outlook and Questions for Later

A) Next step: further separation of VIOLENCE & SCARINESS into physical vs. psychological violence

→ Are there existing taxonomies and annotated data?

B) In development: Reading environment for explicit content detection and 3-level trigger warning





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Appendix

Contributions

01

Sound annotation
framework extension to
US-English romance

- ambient sound
- character sound

02

Scene-level ternary explicit
content assessment for
literary fiction

- VIOLENCE & SCARINESS
- SEX & NUDITY

03

Preliminary empirical
evidence for high *character
sound* loudness SD being a
signal for *erotic* and
violent/scary scenes in
romance fiction

Loudness Level Labeling

S	explicitly no sound	silence (<i>Schweigen</i> and <i>Stille</i>)
1	very low sound	falling leaves – breathing
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Scene Definition

A scene is defined as a segment of the discourse (presentation) of a narrative

- a) when time is – more or less – similar in discourse and histoire,
- b) when place stays – more or less similar,
- c) when it centers around a particular action, and
- d) the character constellation stays – more or less – the same.

Event Type Classification

non_event
stative_event
process_event
change_of_state

The event type <**change_of_state**> covers physical and mental state changes of animate and inanimate entities. The change of state needs to be expressed in a single verbal phrase. An example of a (mental) change of state is the process of waking up, described by the first verbal phrase in Example 1.

The event type <**process_event**> covers actions and happenings that do not lead to a change of state, such as processes of moving, talking, thinking and feeling. For example, the second verbal phrase in Example 1 is a process of perception (“found [...] himself [...] transformed”) and, therefore, has to be annotated as a process event.³

The event type <**stative_event**> covers all verbal phrases that refer to the state of an animate or inanimate entity. This also includes physical and mental states. Unlike changes of state and processes, stative events do never refer to temporal processes. As Example 2 shows, stative events are often descriptions of places, persons or objects.

2. [₁There stood a bowl filled with sweet milk]₁ [₂in which swam small bits of white bread.]₂

The event type <**non_event**> finally covers verbal phrases that do not refer to a fact in the narrated world. The most common variants of non-events are questions or modalized and generic statements (Example 3). Non-events are a necessary addition allowing us to annotate all text passages not just those referring to events, thereby simplifying the annotation schema.

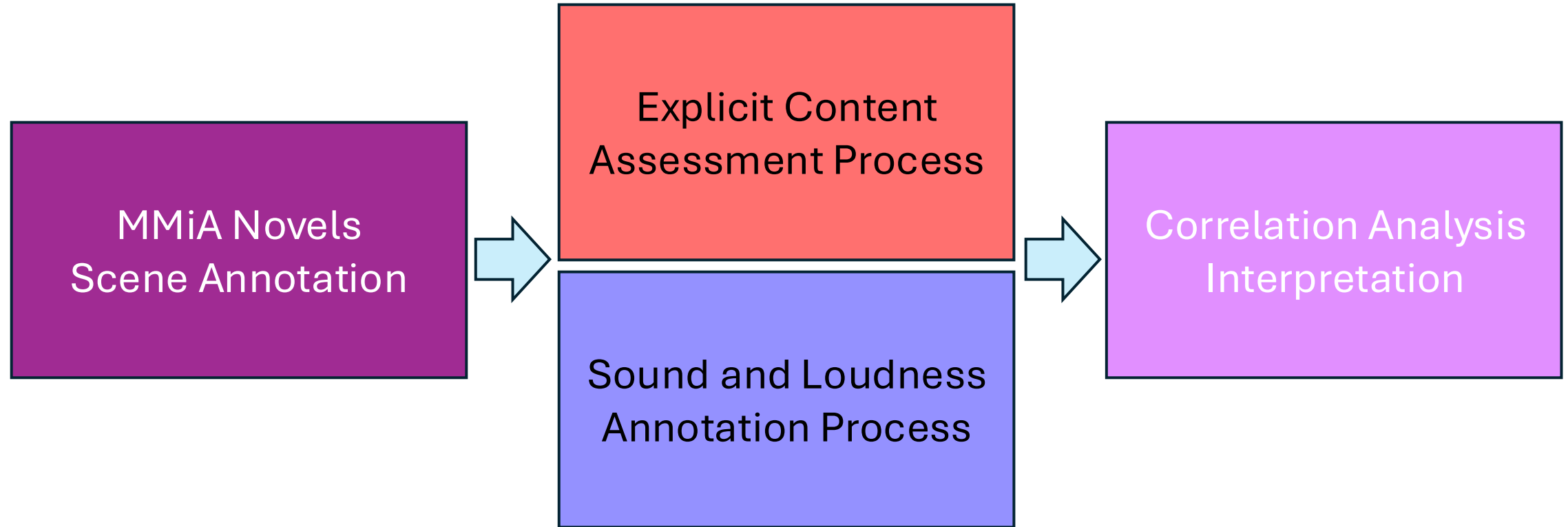
3. [₁A man must have his sleep.]₁

Vauth et al. (2021: 336)

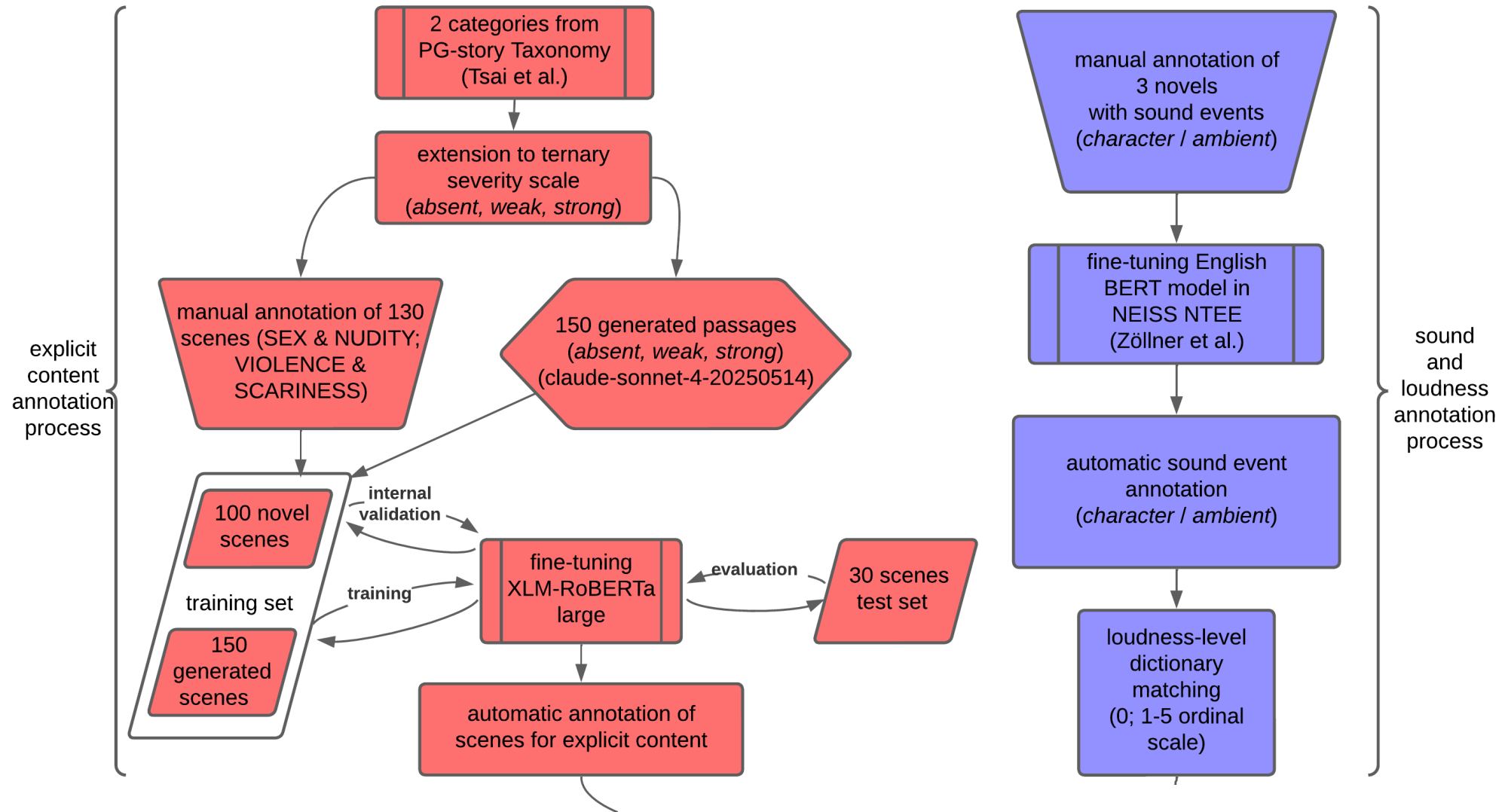
Research on Scene Segmentation

- Evelyn Gius, et al. 2019. **Detection of Scenes in Fiction**. In Book of Abstracts, Utrecht.
- Evelyn Gius, et al. 2021. **Guidelines for Detection of Scenes**. Zenodo.
- Albin Zehe, et al. 2021a. **Detecting Scenes in Fiction: A new Segmentation Task**. In Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume, p. 3167–3177, Online. Association for Computational Linguistics.
- Albin Zehe, et al., editors. 2021b. **Proceedings of the Shared Task on Scene Segmentation**, Vol. 3001 of CEUR Workshop Proceedings. CEUR, KONVENS 2021, Düsseldorf.
- Murathan Kurfalı and Mats Wirén. 2021. “**Breaking the Narrative: Scene Segmentation through Sequential Sentence Classification**.” In Proceedings of the Shared Task on Scene Segmentation, 49-53, 2021.
- Albin Zehe, et al. 2025. **Assessing the state of the art in scene segmentation**. Proceedings of the NAACL 2025.
- Svenja Guhr, et al. 2025. **Rethinking Scene Segmentation. Advancing Automated Detection of Scene Changes in Literary Texts**. Proceedings of the LaTeCH SIGHUM Workshop at NAACL 2025.

Many New Terms and Concepts



Parallel Workflow



XLM RoBERTa

- The SEX & NUDITY classifier achieved good results on the external test set (macro F1 = 0.63) even under low-resource conditions.
- The VIOLENCE & SCARINESS classifier (macro F1 = 0.30) proved harder to generalize, likely because the category bundles narratologically distinct phenomena: violence and scariness

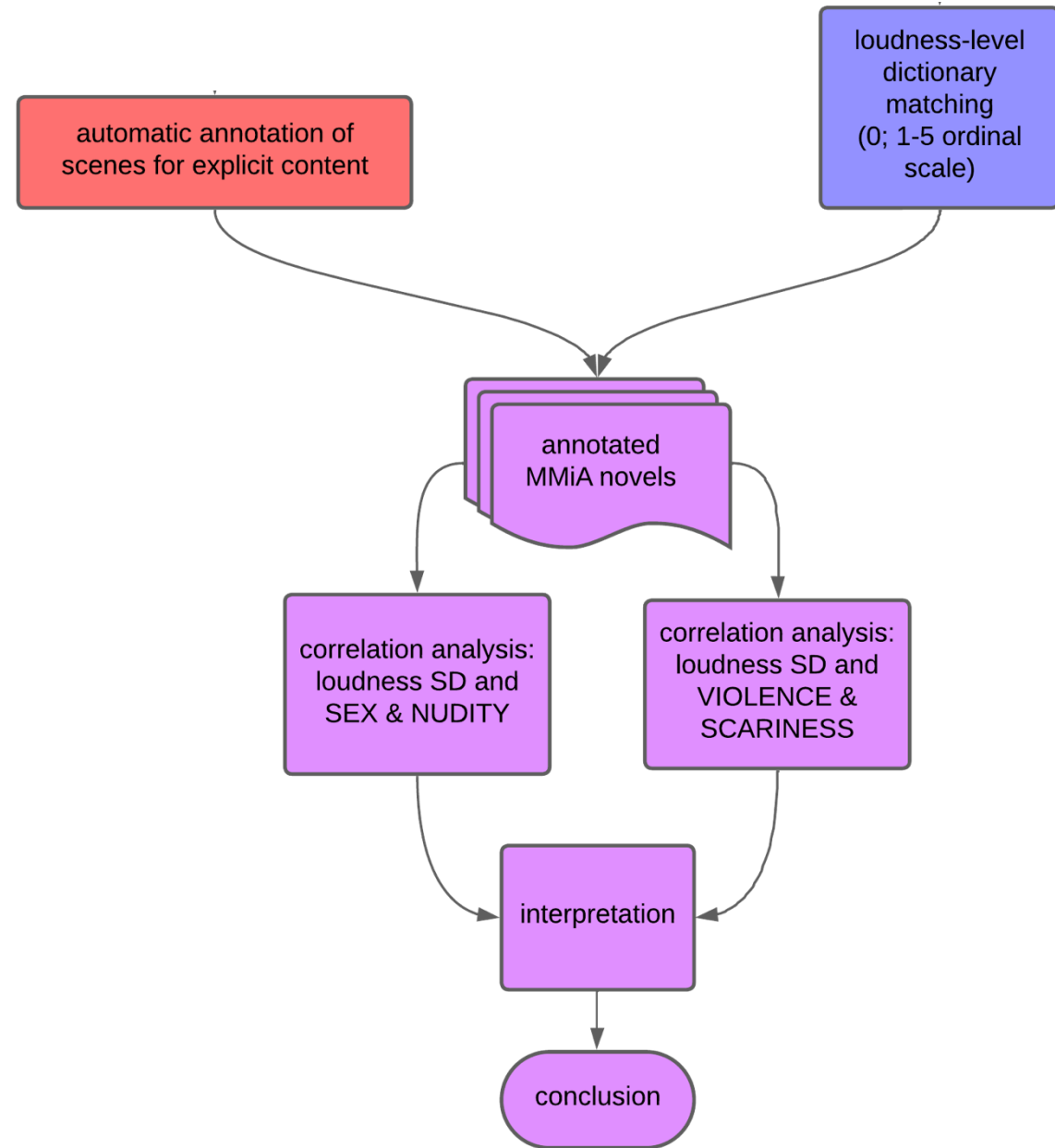
Class	P	R	F1	Sup
<i>absent</i>	0.70	0.94	0.80	17
<i>weak</i>	0.75	0.30	0.43	10
<i>strong</i>	0.67	0.67	0.67	3
macro avg	0.70	0.64	0.63	30
weighted avg	0.71	0.70	0.66	30

Table 4: SEX & NUDITY: external test ($n = 30$).

Class	P	R	F1	Sup
<i>absent</i>	0.46	0.93	0.62	14
<i>weak</i>	0.00	0.00	0.00	10
<i>strong</i>	1.00	0.17	0.29	6
macro avg	0.49	0.37	0.30	30
weighted avg	0.42	0.47	0.35	30

Table 6: VIOLENCE & SCARINESS: external test ($n = 30$).

Correlation Analysis



Results (1/2)

- *Character sound* loudness SD is an acoustically grounded proxy for explicit content in romance fiction.
- **H2 confirmed:**
 - SD correlates positively with erotic scene score ($p = 0.168$, $p < 0.001$), rising from 0.467 (absent) to 0.644 (strong, +37.9%; AUC = 0.658, $p < 0.001$).

Results (2/2)

- *Character sound* SD is elevated in both categories, suggesting it may signal narrative intensity more broadly, not eroticism alone.
- **H1 inconclusive:**
 - Violent/scary scenes show higher character SD ($M = 0.749$ vs. 0.509 , $p = 0.004$), ambient SD returns null results ($p = 0.661$), limited by sparse annotation.



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